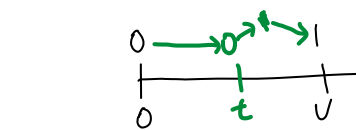
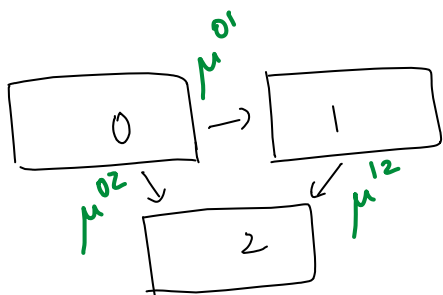


μ_{x+t}^{ij} force of transition
transition intensity

Assumption 3

For all states i and j , and all ages $x \geq 0$,
we assume ${}_t p_x^{ij}$ is a differentiable function of t .
- smooth so that the limit exists

§.4



$${}_u p_x^{01} = \int_0^u {}_t p_x^{\overline{00}} \cdot \mu_{x+t}^{01} \cdot {}_{u-t} p_{x+t}^{\overline{11}} dt$$

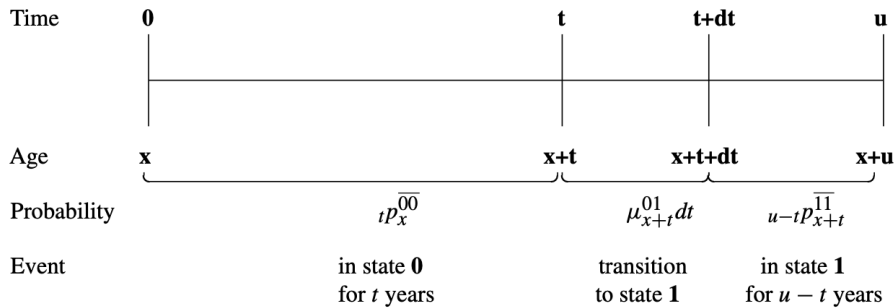
$${}_t p_x^{\overline{ii}} = e^{-\int_0^t \sum_{\substack{j=0 \\ j \neq i}}^n \mu_{x+s}^{ij} ds}$$

Resist μ 's
which could
pull them out
of state i

For any state i in a multiple state model,

$${}_t p_x^{ii} = \exp \left\{ - \int_0^t \sum_{j=0, j \neq i}^n \mu_{x+s}^{ij} ds \right\}.$$

LTAM ${}_t p_x = \bar{e}^{\int_0^t \mu_{x+s} ds}$
 Constant μ ${}_t p_x = \bar{e}^{\mu t}$



$${}_u p_x^{01} = \int_0^u {}_t p_x^{00} \mu_{x+t}^{01} {}_{u-t} p_{x+t}^{11} dt.$$

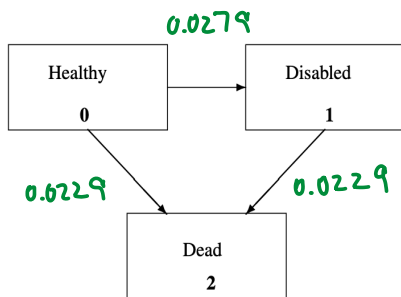


Figure 8.3 The permanent disability model.

Handwritten calculation for ${}_{10}P_{60}^{02}$ using transition intensities:

$$\begin{pmatrix} {}_{10}P_{60}^{00} & {}_{10}P_{60}^{01} & {}_{10}P_{60}^{02} \\ {}_{10}P_{60}^{10} & {}_{10}P_{60}^{11} & {}_{10}P_{60}^{12} \\ {}_{10}P_{60}^{20} & {}_{10}P_{60}^{21} & {}_{10}P_{60}^{22} \end{pmatrix} = \begin{pmatrix} 0.6017 & 0.1936 & 0.2047 \\ 0 & 0.7953 & 0.2047 \\ 0 & 0 & 1 \end{pmatrix}$$

Example 8.4 Consider the permanent disability model illustrated in Figure 8.3.

(a) Suppose the transition intensities for this model are all constants, as follows

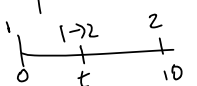
$$\mu_x^{01} = 0.0279, \quad \mu_x^{02} = 0.0229, \quad \mu_x^{12} = \mu_x^{02}.$$

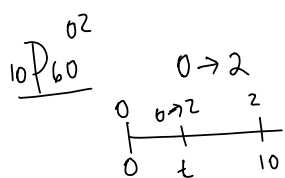
Calculate ${}_{10}P_{60}^{00}$ and ${}_{10}P_{60}^{01}$.

Handwritten calculations:

$$\begin{aligned}
 {}_{10}P_{60}^{00} &= {}_{10}P_{60}^{\bar{00}} = e^{-\int_0^{10} (0.0279 + 0.0229) dt} = e^{-0.508(10)} = 0.6017 \\
 {}_{10}P_{60}^{01} &= \int_0^{10} e^{-(0.0279 + 0.0229)t} \cdot (0.0279) \cdot e^{-0.0229(10-t)} dt \\
 &= \int_0^{10} e^{-0.508t} (0.0279) e^{-0.0229(10)} e^{0.0229t} dt \\
 &= (0.0279) e^{-0.0229(10)} \int_0^{10} e^{-0.0279t} dt = 0.1936 \\
 {}_{10}P_{60}^{11} &= {}_{10}P_{60}^{\bar{11}} = e^{-0.0229(10)} = 0.7953
 \end{aligned}$$

$${}_{10}p_{60}^{12} = \int_0^{10} e^{-0.0229t} (0.0229) \cdot 1 dt = \underline{0.2047}$$

$$= 0.0229 \left. \frac{e^{-0.0229t}}{-0.0229} \right|_0^{10}$$




$$\int_0^{10} e^{-(0.0279 + 0.0229)t} \cdot (0.0229) (1) dt = \underline{0.1795}$$

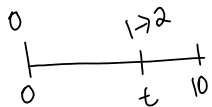
+ 0.2047

Two different methods to solve

Negation

① $\int_0^{10} e^{-0.0508t} \cdot (0.0279) (1 - e^{-0.0229(10-t)}) dt = \underline{0.0251}$

not staying in state 1 = moving to state 2

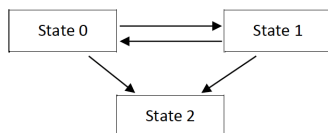


② $\int_0^{10} {}_t p_x^{01} \cdot \mu_{x+t}^{12} dt$

Need a function for ${}_t p_x^{01}$

LTAM

8.11. A continuous Markov process is modeled by the following multiple state diagram:



You are given the following constant transition intensities:

(i) $\mu^{01} = 0.08$

(ii) $\mu^{02} = 0.04$

(iii) $\mu^{10} = 0.10$

(iv) $\mu^{12} = 0.05$



For a person in State 1, calculate the probability that the person will continuously remain in State 1 for the next 15 years.

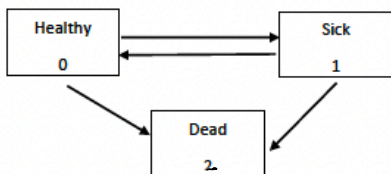
- (A) 0.032
 (B) 0.105
 (C) 0.151
 (D) 0.250
 (E) 0.350

$${}_{15}P_x = e^{-\int_0^{15} (0.10 + 0.05) ds} = e^{-0.15(15)} = 0.1054$$

$$e^{-(0.10 + 0.05 + 0.08) \cdot 15} = 0.032$$

LTAM

8.20. A 5-year sickness insurance policy is based on the following Markov model:



You are given the following constant forces of transition:

(i) $\mu^{01} = 0.05$ 01

(ii) $\mu^{10} = 0.02$ 10

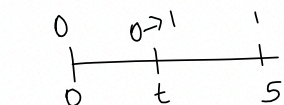
(iii) $\mu^{02} = 0.01$ 02

(iv) $\mu^{12} = 0.06$ 12

There is no notation for this (and that's okay)

Calculate the probability that a Healthy life will become Sick exactly once during the 5 years and remain continuously Sick from that point until the end of the 5 years.

- (A) 0.06
 (B) 0.09
 (C) 0.12
 (D) 0.15
 (E) 0.18



$$\int_0^5 e^{-(0.05 + 0.01)t} (0.05) e^{-(0.02 + 0.06)(5-t)} dt = 0.1762$$

${}^5P_x^{01}$
 $0 \rightarrow 1$
 $0 \rightarrow 1 \rightarrow 0 \rightarrow 1$
 $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1$
 $\vdots$
 infinity